

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL1- INDUSTRY PROBLEM TASK

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 4

Time Duration : 1 Hour
Total: Marks:40

Code of Conduct:

I (K.B.Devarshan/192511426) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach & Feature Selection

- **Approach:** The suitable approach is **Supervised Classification Learning**. We are given historical patient records with predefined outcomes (labels) and aim to induce a general set of rules (a model) that maps patient features to a specific diagnostic category.
- **Target Variable:** Diabetes_Diagnosis (Binary: Yes/No).
- **Key Features (with clinical justification):**
 1. **Glucose Level:** The most direct physiological indicator of diabetes. High fasting blood sugar is the primary diagnostic criterion.
 2. **BMI (Body Mass Index):** Obesity heavily impacts insulin resistance, making it a critical predictor for Type 2 diabetes.
 3. **Family History:** Genetics play a massive role in predisposition; having a diabetic parent significantly increases risk.

(b) Handling Class Imbalance

Medical datasets typically feature a "class imbalance," where the healthy population (majority) vastly outnumbers the diagnosed population (minority).

- **Strategy: SMOTE (Synthetic Minority Over-sampling Technique).**
- **Justification:** Under-sampling would discard valuable patient data, potentially losing critical patterns of the majority class. Class weighting is a good alternative, but SMOTE explicitly generates synthetic, plausible diabetic patient profiles by interpolating between existing minority instances. This expands the decision boundary for the minority class without perfectly replicating data (which causes overfitting).

(c) Data Splitting and Preprocessing

- **Split Ratio: 70:30 (Train:Test).**
 - *Justification:* In medical use cases, model generalizability is a matter of patient safety. A 30% holdout provides a large enough unseen sample to rigorously evaluate metrics like Recall (sensitivity) and ensures the model isn't just memorizing the training data.
- **Preprocessing Steps:**
 1. **Normalization (Min-Max Scaling):** Applied to continuous features (Glucose, BMI, Blood Pressure). *Impact:* Ensures that variables with large ranges (e.g., Glucose 70-200) don't mathematically dominate variables with smaller ranges (e.g., BMI 18-40) in distance-based or gradient-based algorithms.
 2. **Encoding:** One-Hot Encoding for nominal categorical data and Label Encoding for ordinal data. *Impact:* Converts text (e.g., Family History: Yes/No) into a machine-readable numeric format without falsely implying a mathematical ranking.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree Construction

When constructing the tree, nodes are selected based on the highest Information Gain or lowest Gini Impurity.

- **Root Node Justification: Glucose Level** is chosen as the root node. Clinically and mathematically, it provides the highest **Information Gain**:

$$IG(S, A) = H(S) - \sum \frac{|S_v|}{|S|} H(S_v)$$

Splitting the dataset by Glucose > 125 mg/dL creates the purest immediate child nodes compared to splitting by BMI or Age.

- **Illustrative 3-Level Structure:**
 - **Level 1 (Root):** Glucose > 125?
 - *If Yes:* Go to Level 2A.
 - *If No:* Go to Level 2B.
 - **Level 2A:** BMI > 30?
 - *If Yes:* Predict **Diabetic** (Leaf).
 - *If No:* Go to Level 3A (Family History?).
 - **Level 2B:** Age > 50?
 - *If Yes:* Go to Level 3B (Blood Pressure?).
 - *If No:* Predict **Non-Diabetic** (Leaf).

(b) Model Evaluation & False Negatives

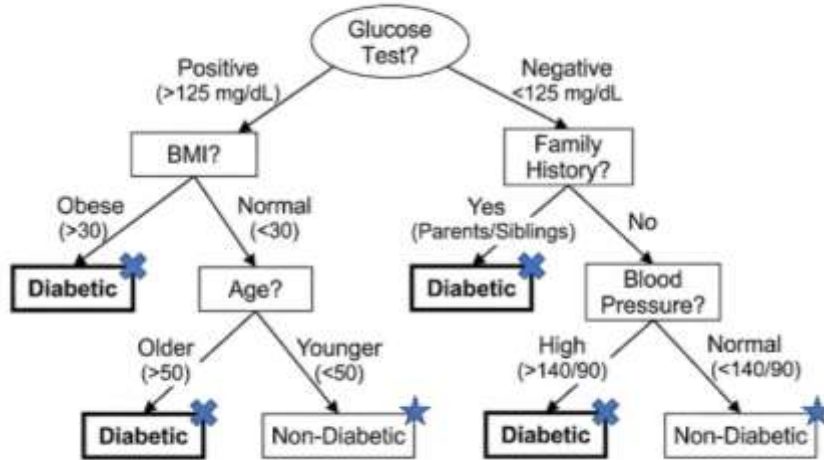
- **Illustrative Metrics:** Accuracy: 88%, Precision: 82%, Recall: 94%, F1-Score: 87%.
- **False Negatives (Missed Cases):** A False Negative occurs when the model predicts "Non-Diabetic," but the patient actually has diabetes. In healthcare, this is the most dangerous error because a sick patient will go untreated.
- **How to Reduce FNs:** We must optimize for **Recall (Sensitivity)**. This can be done by lowering the classification probability threshold (e.g., predicting "Diabetic" if probability > 0.35 instead of 0.50) or by assigning a higher penalty weight to False Negatives in the model's loss function.

(c) Post-Pruning

- **Concept:** Post-pruning (Cost-Complexity Pruning) grows a massive, overfitted tree, then systematically snips off leaf nodes that do not significantly improve validation accuracy, controlled by a complexity parameter α .
- **Comparison:** Pre-pruning (stopping early) can suffer from the "horizon effect," missing complex multi-feature interactions deeper in the tree. Post-pruning ensures we see the full picture before simplifying.
- **Clinical Deployment: Post-pruning is highly preferable.** It initially captures complex medical interactions (e.g., the combined effect of low BMI but high genetic risk) but

removes noise, resulting in a model that is both highly accurate and generalizable to new patients.

DIABETES DECISION TREE (Illustrative) - Digital Edition
Course CSA1709: Artificial Intelligence / Task 2: Decision Tree for Diagnosis



Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression vs Decision Tree

- **Performance Comparison:** Logistic Regression (LR) often achieves a similar Accuracy to DTs but usually yields a superior **AUC-ROC score** (e.g., 0.91 vs 0.86).
- **When LR is Preferable:** LR is a statistical model that outputs continuous probabilities (e.g., "This patient has a 72% probability of developing diabetes"). This is crucial for **Risk Stratification**, allowing doctors to triage patients into Low, Medium, and High-risk groups rather than a rigid Yes/No output. LR is preferable when the relationships between features and log-odds are relatively linear.

(b) Feature Importance Analysis

- **Decision Tree Top 3:** 1. Glucose, 2. BMI, 3. Age. (Derived from Gini Importance / total reduction of impurity).
- **Logistic Regression Top 3:** 1. Glucose, 2. Family History, 3. BMI. (Derived from the absolute magnitude of the learned coefficients).
- **Do they agree?** Mostly, yes. Both models identify Glucose and BMI as top predictors. However, LR might rank Family History higher if it has a strong linear correlation, whereas the DT might rank Age higher if Age acts as a great "splitting" threshold for multiple subgroups.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP (Markov Decision Process) Formulation

To train an RL agent to prescribe treatments, we define the MDP tuple (S, A, P, R) :

- **States (S):** Patient health stages defined by clinical markers.
 $\{S_0:\text{Healthy}, S_1:\text{Pre-diabetic}, S_2:\text{Mild Diabetes}, S_3:\text{Severe Diabetes}\}.$
- **Actions (A):** $\{A_1:\text{Monitor}, A_2:\text{Diet}, A_3:\text{Exercise}, A_4:\text{Medication}\}.$

- **Reward Function (R):**
 - +10 for transitioning to a healthier state (e.g., $S_2 \rightarrow S_1$).
 - -10 for degrading to a worse state.
 - -1 penalty for aggressive actions (Medication) to encourage lifestyle interventions (Diet/Exercise) when possible.
- **Transition Dynamics (P):** The probability $P(s'|s, a)$ of moving to next state s' given action a in state s . (e.g., Diet applied to Pre-diabetic has an 80% chance to transition to Healthy).

(b) Q-Learning Application

The agent uses the Bellman equation to update the Q-table:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[R + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

(Assuming Learning Rate $\alpha = 0.1$, Discount Factor $\gamma = 0.9$)

Illustrative Q-Table Updates (Iteration 1 & 2):

- **Step 1:** Patient is in S_1 (Pre-diabetic). Agent chooses A_2 (Diet). Patient transitions to S_0 (Healthy). Reward = +10.

$$\text{Update: } Q(S_1, \text{Diet}) = 0 + 0.1[10 + 0.9(0) - 0] = 1.0$$

- **Step 2:** Patient is in S_2 (Mild). Agent chooses A_1 (Monitor). Patient degrades to S_3 (Severe). Reward = -10.

$$\text{Update: } Q(S_2, \text{Monitor}) = 0 + 0.1[-10 + 0.9(0) - 0] = -1.0$$

- **Step 3 (Iteration 2):** Patient in S_1 . Agent chooses A_2 (Diet). Transitions to S_0 . Reward = +10.

$$\text{Update: } Q(S_1, \text{Diet}) = 1.0 + 0.1[10 + 0.9(0) - 1.0] = 1.9$$

Convergence Criterion: Training stops when the maximum change in any Q-value across all state-action pairs in an episode falls below a very small threshold ϵ (e.g., 0.001), indicating the policy has stabilized.

(c) Extracted Policy & Ethical Considerations

- **Learned Policy:** State $S_1 \rightarrow$ Diet/Exercise; State $S_2 \rightarrow$ Medication; State $S_0 \rightarrow$ Monitor.
- **Ethical Considerations:**
 1. **Patient Safety (Exploration vs Exploitation):** An RL agent learns by trial and error (exploration). In reality, we cannot "experiment" with denying medication to a Severe diabetic just to see what happens. RL in healthcare must be trained on offline historical data, not live patients.
 2. **Algorithmic Bias:** If the historical transition dynamics (P) heavily feature one demographic, the recommended policy might be ineffective or harmful to minorities with different physiological responses to medication.
 3. **Explainability:** Q-learning provides a matrix of numbers, which is a "black box" to doctors. Medical professionals require explainable AI to trust recommendations; they need to know *why* the agent suggested Medication over Diet for a specific borderline case.